# Constructive and Destructive Training Regimes: Gradient Diagnostics for Neural Network Stability

Thiago Mattar, Cristiano L. Castro, Frederico Coelho, Yan G. Ferreira, Antonio P. Braga

***Abstract*—Curvature governs how gradients change along a learning step. We develop an online diagnostic framework that estimates directional curvature from consecutive training gradients, without computing the Hessian or its vector products. Under full-batch gradient descent, the estimator equals directional curvature averaged along a smooth step and approximates curvature at its starting point. It requires one retained gradient vector and linear-time reductions, with no additional backward pass. Adjacent-gradient cosine identifies constructive and destructive regimes; the curvature estimate combines this alignment with gradient-norm growth. A local loss analysis explains how destructive regimes sustain descent and how amplified reversal produces loss-increasing steps. Controlled CIFAR-10 and WikiText-2 experiments characterize these regimes under GD and sampling. Under Adam, the same measurements reveal curvature response along adaptive steps and early signs of developing loss spikes. Across 129 recovered spikes larger than 10%, gradient norms increase in every event, and 117 of 127 defined pre-rise cosines identify a destructive regime. Joint measures of pre-rise alignment and rise norm amplification explain 38–94% of the within-run variation in log relative peak loss. The framework makes curvature-related instability observable during training and provides a basis for interpreting stabilization heuristics and designing diagnostic-guided interventions.**

***Index Terms*—Training Stability, Directional Curvature, Gradient Alignment, Stochastic Gradient Descent, Adam**

## I. Introduction

Neural networks often reduce training loss while successive gradients point in nearly opposite directions. This behavior becomes especially visible at large learning rates, where rapid overall progress can coexist with oscillation and sharp loss spikes [1]–[3]. For practitioners, the important question is when such oscillation remains productive and when it develops into a damaging excursion. Answering it requires diagnostics that expose the changing training dynamics early enough to inform a response.

We approach this question through *constructive and destructive regimes*. The cosine similarity $\rho_t$ between adjacent raw gradients identifies the regime: positive values indicate directional reinforcement, and negative values indicate directional opposition. The gradient norm supplies the complementary information about the strength of that interaction. Together, they distinguish persistent reversal during descent from reversal accompanied by rapid gradient amplification. This distinction is central to the paper: a destructive regime can sustain loss reduction, while sufficiently strong amplification can turn its local oscillations into loss-increasing steps.

Classical stability analysis relates step size to the Hessian spectrum, and stochastic extensions incorporate variation induced by sampling [4], [5]. The Edge-of-Stability literature establishes that deep networks can continue to improve while operating near or beyond the local quadratic stability boundary [1], [6], [7]. Gradient-product diagnostics and hypergradient methods already exploit successive directions to detect optimization phases or adapt the learning rate [8]–[10]. Our contribution is to connect these temporal interactions to a common account of regime, gradient amplification, net parameter motion and loss-spike anatomy, using measurements available along the training trajectory.

The formulation starts from an exact decomposition of SGD displacement into individual step magnitudes and interactions across all temporal lags. Adjacent-gradient cosine describes the current regime; normalized displacement measures how much traveled distance survives as net motion. A local loss expansion then shows why destructive steps can either decrease or increase loss, depending on gradient-norm growth. The projected-change diagnostic $\eta\widehat{\kappa}_t$ summarizes this interaction and connects it to directional curvature. These quantities require linear-time vector operations on gradients already computed by training, making them suitable for online monitoring.

We first characterize the regimes in controlled full-batch GD and mini-batch SGD experiments on CIFAR-10 and WikiText-2. We then extend the analysis to Adam, whose adaptive directions transform the raw gradient before each update [11]. Raw-gradient diagnostics still expose the loss landscape encountered along that adaptive path: the largest spikes show diagnostic changes before the loss maximum, and the Transformer example reveals a warning before its pronounced rise. Across 129 recovered spikes larger than 10%, gradient norms grow in every event and 117 of the 127 defined pre-rise cosines identify a destructive regime. Combining that initial regime with subsequent norm amplification accounts for a substantial fraction of the variation in spike magnitude.

The paper makes three contributions:

- A unified diagnostic formulation linking constructive and destructive regimes, cumulative parameter displace-

Thiago Mattar, and Yan G. Ferreira are with the Graduate Program in Electrical Engineering, Universidade Federal de Minas Gerais, Belo Horizonte 31270-901, Brazil (e-mail: thiagocmattar@gmail.com; yanvgferreira@gmail.com).

Cristiano L. Castro is with the Graduate Program in Electrical Engineering and the Department of Electrical Engineering, Universidade Federal de Minas Gerais (UFMG), Belo Horizonte, Minas Gerais 31270-901, Brazil (e-mail: crislcastro@gmail.com).

Frederico Coelho, and Antonio P. Braga are with the Graduate Program in Electrical Engineering and the Department of Electronic Engineering, Universidade Federal de Minas Gerais (UFMG), Belo Horizonte, Minas Gerais 35931-008, Brazil (e-mail: fredgfc@ufmg.br; apbraga@ufmg.br).

ment and local loss changes, with explicit curvature and sampling-noise terms.
- A controlled empirical characterization showing how destructive regimes sustain descent, reduce displacement retention and change under mini-batch sampling.
- An extension to Adam that identifies early signs and recurring anatomy of loss spikes through raw-gradient alignment and amplification, supported by individual events, amplitude distributions and quantitative associations.

Section II positions the diagnostics within stability and optimization research. Section III develops the formulation, and Section IV characterizes the regimes under GD and SGD. Section V examines their role in Adam loss spikes and motivates diagnostic-guided interventions. Section VI concludes.

## II. Background and Related Diagnostics

Training dynamics reflect both the geometry encountered by an update and its interaction with preceding updates. Curvature-based analyses explain local amplification; stochastic analyses add sampling variation; temporal diagnostics describe how successive directions reinforce or oppose one another. We use these perspectives to develop a practical account of constructive and destructive regimes.

### A. Curvature, Oscillation and Stability

Let $\theta_t \in \mathbb{R}^p$ denote the parameters after update $t$ and $g_t$ the gradient evaluated at $\theta_{t-1}$. For a constant learning rate $\eta > 0$, plain SGD follows

$$\theta_t = \theta_{t-1} - \eta g_t. \tag{1}$$

In full-batch GD, $g_t = \nabla\mathcal{L}(\theta_{t-1})$ for the training objective $\mathcal{L}$. Writing $H_t = \nabla^2\mathcal{L}(\theta_{t-1})$, the deterministic update has Jacobian

$$J_t = I - \eta H_t. \tag{2}$$

This matrix describes the response to a small parameter perturbation. For a fixed positive-definite quadratic Hessian, each eigencomponent is multiplied by $1 - \eta\lambda_i$, and linear stability requires

$$\max_i |1 - \eta\lambda_i| < 1 \quad \Longleftrightarrow \quad \eta\lambda_{\max}(H) < 2. \tag{3}$$

The component contracts without changing sign when $0 < \eta\lambda_i < 1$, alternates while contracting when $1 < \eta\lambda_i < 2$, and alternates with increasing magnitude when $\eta\lambda_i > 2$ [4], [12]. Thus oscillation and amplification are distinct features even in the simplest local model.

Deep-network trajectories repeatedly enter oscillatory regions. Cohen et al. [1] show that full-batch GD often approaches $\lambda_{\max} \simeq 2/\eta$ while reducing loss over longer timescales. Analyses of this Edge-of-Stability behavior explain how nonlinear dynamics sustain progress near the quadratic boundary [6], [7]. The trajectory measurements of Xing et al. [13] likewise describe steps crossing between sides of a valley. These results motivate examining the gradients produced by successive crossings: their alignment reveals reversal, and their magnitudes reveal whether that reversal is shrinking or growing.

### B. Sampling Noise and Local Geometry

Mini-batch sampling changes both the estimated direction and the trajectory followed through the objective [14]–[16]. Under fresh, conditionally unbiased sampling, write

$$\begin{aligned} g_t &= \nabla\mathcal{L}(\theta_{t-1}) + \xi_t, \\ \mathbb{E}[\xi_t \mid \theta_{t-1}] &= 0, \\ \mathrm{Cov}(\xi_t \mid \theta_{t-1}) &= \frac{N-b}{b(N-1)}\Sigma(\theta_{t-1}), \end{aligned} \tag{4}$$

where the last equality assumes a uniformly sampled size-$b$ subset of $N > 1$ examples without replacement. At a fixed state, let $h_i = \nabla\ell_i(\theta)$ and $\mu = N^{-1}\sum_i h_i = \nabla\mathcal{L}(\theta)$. The per-example covariance is

$$\Sigma(\theta) = \frac{1}{N}\sum_{i=1}^{N}(h_i - \mu)(h_i - \mu)^\top. \tag{5}$$

Its trace measures the total squared dispersion around the full gradient. The finite-population factor in Eq. (4) shows how larger batches reduce this dispersion.

Curvature and stochasticity also interact. The break-even analysis of Jastrzebski et al. [5] links the early learning rate and batch size to the curvature and gradient variability encountered later. In its local one-dimensional quadratic model, variation in sampled curvature adds to the mean-square amplification produced by the mean curvature. This gives a reason to examine the regime along the evolving path as well as the sampling variance at a fixed point. Section III-B makes that distinction explicit for adjacent-gradient alignment.

### C. Temporal Gradient Diagnostics

Successive gradient products have a substantial history in optimization diagnostics. Chee and Toulis [8] use them to detect the transition toward a stationary phase of constant-step SGD. Pesme et al. [9] analyze the classical gradient-product test and show how its small signal relative to variation can impair phase detection. Baydin et al. [10] derive online learning-rate adaptation from hypergradients; for ordinary gradient descent, the resulting update uses the inner product of successive gradients.

Our focus is the anatomy of the regime that these temporal interactions reveal. We combine adjacent alignment with gradient amplification and cumulative displacement, then connect those measurements to loss behavior. This joint view explains how destructive regimes support sustained descent and identifies the additional amplification associated with large loss excursions. Extending the same measurements to raw gradients under Adam provides a common interpretation across plain and adaptive updates.

### D. Loss Spikes in Practical Training

Large-model training reports show why early diagnostics matter. PaLM describes approximately 20 loss spikes during 540B-model training and recovers by restarting from an earlier checkpoint and skipping batches [2]. OPT reports growing activation norms and loss divergence, with recovery through checkpoint restarts and learning-rate reduction; tighter gradient

clipping also improves early stability [17]. These interventions incur training cost and make the timing of a warning practically consequential.

Modern recipes therefore monitor gradient norms alongside loss. OLMo 2 uses rolling loss and gradient-norm spike scores in its stability analysis [3], while Llama 3 emphasizes numerical precision, warmup and batch-size scheduling in a largely stable pretraining recipe [18]. Adjacent-gradient cosine adds a directional interpretation to such magnitude measurements: it identifies whether a growing gradient reinforces or opposes the preceding one. The following section develops this combination into diagnostics of constructive and destructive training.

## III. Characterizing Constructive and Destructive Regimes

Our diagnostics answer three connected questions: which directional regime is the optimizer traversing, how strongly are gradients changing, and how much of the resulting motion accumulates? Adjacent-gradient cosine $\rho_t$, the projected change $\eta\widehat{\kappa}_t$, and normalized displacement $R_T$ supply these local and cumulative views. We first derive them for plain SGD, then explain how destructive regimes can sustain descent or develop loss-increasing steps.

Consider the plain SGD update in Eq. (1), with constant learning rate $\eta > 0$. As before, $\theta_t \in \mathbb{R}^p$ denotes the parameters after update $t$, and $g_t$ is the mean mini-batch gradient evaluated at $\theta_{t-1}$. Summing the updates gives the displacement from initialization,

$$\Delta_T := \theta_T - \theta_0 = -\eta \sum_{t=1}^{T} g_t. \tag{6}$$

The displacement is the net parameter change produced by $T$ updates. Expanding its squared norm separates individual step magnitudes from interactions between steps:

$$\begin{aligned} \|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 \\ + 2\eta^2 \sum_{1 \le u < t \le T} g_u^\top g_t. \end{aligned} \tag{7}$$

The terms $\|g_t\|^2$ are nonnegative and determine the squared lengths of individual SGD steps through $\eta^2\|g_t\|^2$. In full-batch GD, a larger gradient norm indicates a steeper local loss and, therefore, a larger first-order decrease along the negative gradient. The cross-step terms $g_u^\top g_t$ describe how these steps interact – positive inner products reinforce the displacement, whereas negative inner products cancel part of it. This decomposition provides the geometric basis for interpreting reinforcement and cancellation along a training trajectory.

To isolate directional alignment from gradient magnitudes, define the cosine similarity between any two nonzero gradients and its consecutive-step form

$$\begin{aligned} \rho_{u,t} &:= \frac{g_u^\top g_t}{\|g_u\|\|g_t\|}, \quad 1 \le u < t \le T, \\ \rho_t &:= \rho_{t-1,t}, \qquad t = 2, \dots, T. \end{aligned} \tag{8}$$

We call the adjacent-gradient regime *constructive* when $\rho_t > 0$ and *destructive* when $\rho_t < 0$. Zero denotes orthogonal gradients; a zero gradient norm leaves its cosine undefined. Persistent signs identify sustained regimes, while sign changes mark transitions between them. These definitions refer to raw gradients throughout the paper; under plain SGD, their signs also describe the interaction between consecutive updates.

Substituting Eq. (8) into Eq. (7) gives

$$\begin{aligned} \|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 \\ + 2\eta^2 \sum_{1 \le u < t \le T} \|g_u\|\|g_t\|\rho_{u,t}. \end{aligned} \tag{9}$$

The exact sum retains interactions across every temporal lag. For given gradient magnitudes, positive pairwise terms reinforce net motion and negative terms cancel it. The adjacent statistic $\rho_t$ makes the current regime observable online; longer-lag interactions determine how that local behavior contributes to motion over the full training horizon.

### A. Directional Curvature and Consecutive Alignment

For mini-batch SGD, we explicitly incorporate stochastic gradient noise in Eq. (4), $g_t = \nabla\mathcal{L}(\theta_{t-1}) + \xi_t$. Suppose the full objective $\mathcal{L}$ is twice continuously differentiable along the preceding step, and write $H_{t-1} := \nabla^2\mathcal{L}(\theta_{t-2})$. Expanding the full gradient around $\theta_{t-2}$ and substituting the preceding SGD update yields

$$g_t = (I - \eta H_{t-1})g_{t-1} + (\xi_t - \xi_{t-1}) + r_t, \tag{10}$$

where $r_t$ is the Taylor remainder. Neglecting $r_t$ recovers the first-order approximation. Taking the inner product with $g_{t-1}$ then gives

$$\begin{aligned} \|g_{t-1}\|\|g_t\|\rho_t \approx \underbrace{\|g_{t-1}\|^2}_{\text{Gradient magnitude}} \\ \underbrace{- \eta g_{t-1}^\top H_{t-1} g_{t-1}}_{\text{Curvature contribution}} \\ \underbrace{+ g_{t-1}^\top(\xi_t - \xi_{t-1})}_{\text{Noise contribution}}. \end{aligned} \tag{11}$$

Positive curvature reduces the component of the next gradient along the preceding direction. Its strength relative to the preceding squared norm is the directional curvature

$$\kappa_t := \frac{g_{t-1}^\top H_{t-1} g_{t-1}}{\|g_{t-1}\|^2}, \tag{12}$$

Thus, $\kappa_t$ measures how strongly a step changes the gradient along the direction actually followed, whereas $\rho_t$ measures the resulting alignment. Sampling modifies this response through the normalized noise interaction

$$\varepsilon_t := \frac{g_{t-1}^\top(\xi_t - \xi_{t-1})}{\|g_{t-1}\|^2}. \tag{13}$$

Dividing Eq. (11) by the two gradient norms gives

$$\rho_t \approx \frac{\|g_{t-1}\|}{\|g_t\|}(1 - \eta\kappa_t + \varepsilon_t). \tag{14}$$

In full-batch GD, $\varepsilon_t = 0$. Within this first-order approximation, $\eta\kappa_t < 1$ retains a positive projection, while $\eta\kappa_t > 1$ reverses it: the next update partially opposes the preceding step. At $\eta\kappa_t = 1$, the new gradient has lost its component along the old direction. It is orthogonal if nonzero; if it vanishes, cosine is undefined. The norm ratio determines how this projected change translates into the magnitude of $\rho_t$.

The threshold at one also has an optimization interpretation. Along the full-gradient direction, the local quadratic loss model is $\mathcal{L}(\theta_{t-2} - \alpha g_{t-1}) \approx \mathcal{L}(\theta_{t-2}) - \alpha\|g_{t-1}\|^2 + \alpha^2\kappa_t\|g_{t-1}\|^2/2$. For $\kappa_t > 0$, its directional minimum is at $\alpha = 1/\kappa_t$. Negative alignment therefore identifies a step beyond this directional minimum under the approximation. This is the geometric meaning of the destructive regime: the next gradient opposes a component of the preceding step. The loss-increase boundary depends additionally on the strength of that opposition, as derived in Section III-D.

This interpretation leads to a measurable diagnostic. Rearranging Eq. (14) gives

$$\begin{aligned}\eta\widehat{\kappa}_t &:= 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|} \\ &= -\frac{g_{t-1}^\top (g_t - g_{t-1})}{\|g_{t-1}\|^2} \approx \eta\kappa_t - \varepsilon_t.\end{aligned} \tag{15}$$

This diagnostic measures the change of gradient projected onto its preceding direction, normalized by the preceding squared norm. It combines the regime with the relative gradient magnitude in one observable quantity. In full GD, a small Taylor remainder gives a directional-curvature estimate; under SGD, the explicit noise term describes the sampling contribution. The diagnostic concerns the direction actually traversed, complementing the largest-eigenvalue criterion in Eq. (3).

### B. Noise Contribution under Mini-Batch Sampling

Sampling contributes to the regime through its interaction with the preceding noisy gradient. Write $g_{t-1} = \nabla\mathcal{L}(\theta_{t-2}) + \xi_{t-1}$. Fresh, conditionally unbiased batches give $\mathbb{E}[g_{t-1}^\top \xi_t] = 0$ and $\mathbb{E}[\nabla\mathcal{L}(\theta_{t-2})^\top \xi_{t-1}] = 0$. With finite second moments, the preceding gradient contributes its own noise energy:

$$\mathbb{E}[g_{t-1}^\top(\xi_t - \xi_{t-1})] = -\mathbb{E}[\|\xi_{t-1}\|^2] \le 0. \tag{16}$$

Thus, even unbiased sampling noise contributes a systematic nonpositive term to the expected unnormalized interaction between consecutive gradients.

This contribution is directly related to mini-batch size. Consider a mini-batch of $b$ examples sampled uniformly without replacement from $N > 1$ training examples. Using the per-sample gradient covariance $\Sigma(\theta)$ defined in Eq. (5), the sampling variance satisfies

$$\mathbb{E}[\|\xi_{t-1}\|^2] = \frac{N-b}{b(N-1)}\mathbb{E}[\operatorname{tr}\Sigma(\theta_{t-2})]. \tag{17}$$

At fixed parameters, smaller batches produce larger sampling variance and, consequently, a larger negative expected noise contribution. For full-batch gradients ($b = N$), this contribution vanishes.

Noise can weaken directional coherence by enlarging gradient norms relative to their shared signal. For independent estimates at two fixed states, let $\mu, \nu$ be the nonzero full gradients and $v_\mu, v_\nu$ the expected squared noise norms. When sampled norms concentrate,

$$\mathbb{E}[\cos(\mu + \xi, \nu + \zeta)] \approx \frac{\cos(\mu, \nu)}{\sqrt{(1 + v_\mu/\|\mu\|^2)(1 + v_\nu/\|\nu\|^2)}}.$$

This attenuation moves either sign toward zero. At the same fixed state, the common full-gradient direction gives the special case

$$\mathbb{E}[\rho_t] \approx \frac{\|\nabla\mathcal{L}(\theta)\|^2}{\|\nabla\mathcal{L}(\theta)\|^2 + \frac{N-b}{b(N-1)}\operatorname{tr}\Sigma(\theta)}. \tag{18}$$

Here the expected cosine is nonnegative; the approximation quantifies the dilution of a common direction by sampling. During SGD, a sampled update also changes the next state, coupling noise and curvature. The fixed-state attenuation argument therefore isolates a mechanism, while the batch-size intervention in Section IV-D tests its expression along evolving trajectories.

### C. Normalized Displacement and Cumulative Interference

The regime describes adjacent directions; normalized displacement describes their cumulative outcome. It compares the straight-line distance from initialization with the total distance traveled:

$$R_T := \frac{\|\Delta_T\|}{\eta\sum_{t=1}^T \|g_t\|} = \frac{\|\sum_{t=1}^T g_t\|}{\sum_{t=1}^T \|g_t\|}, \qquad 0 \le R_T \le 1, \tag{19}$$

For a nonzero accumulated path length under constant-step plain SGD, the denominator is the total distance traveled in parameter space, while the numerator measures the net displacement from initialization. Thus, $R_T$ quantifies the fraction of accumulated step length retained as net displacement.

Using Eq. (9), this ratio satisfies

$$R_T^2 = \frac{\sum_{t=1}^T \|g_t\|^2 + 2\sum_{1\le u<t\le T}\|g_u\|\|g_t\|\rho_{u,t}}{\left(\sum_{t=1}^T \|g_t\|\right)^2}. \tag{20}$$

Thus $R_T$ incorporates interactions across all temporal lags. When $R_T$ is near one, almost all traveled distance survives as net displacement; when it is near zero, little does. Repeated back-and-forth steps and weakly coherent sampled directions both suggest low retention. For example, equal-length orthogonal updates give $R_T = 1/\sqrt{T}$, whereas equal opposing pairs cancel exactly. We therefore expect small $R_T$ when such behavior persists over a substantial part of an SGD path. Absolute displacement can keep growing while its retained fraction falls. Together, $\rho_t$, gradient norms and $R_T$ describe the local regime, the size of its steps and the fraction retained over time.

These diagnostics are inexpensive in computational structure. For $p$ parameters, a norm and an inner product with the retained preceding gradient give $\rho_t$ and $\eta\widehat{\kappa}_t$ in $O(p)$ operations and $O(p)$ additional storage per update. Computing $R_T$ adds a running vector sum and a scalar sum of norms, also with

$O(p)$ work and storage independent of $T$. All use gradients already produced by the training backward pass; the exact all-pair identity is evaluated through these sums. Explicit lag profiles are a separate measurement requiring retained history.

### D. From Destructive Descent to Loss Increase

A destructive step can still decrease loss after crossing its directional minimum. The decisive quantity is how strongly the new gradient opposes the preceding one. Define the full-objective change of that step as $\delta\mathcal{L}_{t-1} := \mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$. The second-order approximation derived in Appendix A separates three contributions:

$$\begin{aligned}\delta\mathcal{L}_{t-1} \approx\ & \underbrace{-\frac{\eta}{2}\|g_{t-1}\|^2}_{\text{Gradient magnitude}} \\ & \underbrace{-\frac{\eta}{2}\rho_t\|g_{t-1}\|\|g_t\|}_{\text{Directional interference}} \\ & + \underbrace{\frac{\eta}{2}g_{t-1}^\top(\xi_{t-1}+\xi_t)}_{\text{Noise interaction}}.\end{aligned} \tag{21}$$

The magnitude term favors descent. The interference term reinforces it for positive alignment and opposes it for negative alignment: destructive interference contributes positively to the predicted loss change. With negligible noise and nonzero gradients, the approximation predicts a loss increase precisely when

$$\rho_t < -\frac{\|g_{t-1}\|}{\|g_t\|}. \tag{22}$$

The condition requires a destructive regime and sufficient gradient-norm growth: the opposing component of the new gradient must exceed the preceding gradient magnitude. This is the distinction that makes the joint diagnostic useful. The sign identifies the regime, while the norm ratio determines whether its destructive contribution can overwhelm the descent term.

Substituting Eq. (14) with zero noise gives

$$\delta\mathcal{L}_{t-1} \approx \frac{\eta}{2}\|g_{t-1}\|^2(\eta\kappa_t - 2). \tag{23}$$

Within the local quadratic model, $1 < \eta\kappa_t < 2$ is a destructive regime with decreasing loss; $\eta\kappa_t > 2$ produces a loss increase. The projected-change diagnostic expresses the same transition through observable gradient reversal and amplification. Nonlinear remainder and SGD noise terms delimit this approximation. We now test how the predicted regimes appear in neural-network training and how their signatures relate to cumulative motion and loss spikes.

## IV. Constructive and Destructive Regimes in GD and SGD

We first examine whether destructive regimes sustain descent and how their reversals shape cumulative parameter motion. Full-batch GD isolates these dynamics on fixed objectives; matched mini-batch continuations then show how sampling changes the regimes encountered during training.

The comparison covers a tanh MLP on 5,000 CIFAR-10 images, ResNet-32 on all 50,000 CIFAR-10 training images, and a word-level Transformer on WikiText-2. Each minimizes its complete fixed training objective for 3,000 updates at three constant learning rates (Table I). Within each model, rates share one frozen initialization and data preparation; they were selected by earlier training calibrations to expose contrasting dynamics, without test-loss selection. Plain GD uses no momentum, weight decay, clipping or schedule. The three cases test whether the same geometric signatures recur across different objectives and architectures. Appendix B specifies objectives, architectures, precision and measurement details.

### A. Descent in a Destructive Regime

Figure 1 pairs loss with the projected gradient change $\eta\widehat{\kappa}_t$. The higher-rate trajectories develop oscillatory loss while continuing to improve over longer timescales; their late projected change approaches two. Their negative adjacent cosines in Fig. 2 identify a sustained destructive regime. Equation (23) explains why this value is informative: in the local quadratic model, it separates a loss-decreasing step from one that reverses the preceding progress. The observed pattern gives a gradient-based view of the oscillatory descent associated with the Edge of Stability [1].

Consecutive gradients therefore expose an oscillatory regime that is productive over the training horizon. The projected-change diagnostic provides its directional interpretation using inexpensive vector reductions. Its pointwise accuracy depends on the local approximation: at the highest Transformer rate, the unsmoothed noise-free expression in Eq. (21) disagrees with the observed loss-change sign in 44.2% of 2,999 transitions. This finite-step discrepancy qualifies the interpretation of individual updates within the sustained regime.

### B. Gradient Magnitude and Cumulative Motion

Figure 2 explains how the destructive regime changes parameter motion. The lowest MLP and Transformer rates remain constructive, whereas the other seven conditions develop nearly opposing adjacent gradients. In the MLP and ResNet, this transition also accompanies growing gradient norms: the steps become larger even as their directions increasingly cancel. These fluctuations arise without mini-batch sampling.

The decreasing $R_t$ quantifies the cancellation associated with these reversals. A growing fraction of the traveled distance is spent moving back and forth, even as the parameters move farther from initialization. Table I shows the resulting combination: within every model, the highest tested rate has the lowest retention, the largest absolute displacement, and the lowest final training loss. A rate can therefore be large relative to local directional minima yet productive over the full training horizon.

Local destruction coexists with reinforcement over longer horizons. The complete cross-step interaction is positive at the endpoint in all nine trajectories, including the seven with negative accumulated lag-one interaction in Eq. (7). Longer-range reinforcement can coexist with repeated local reversal.

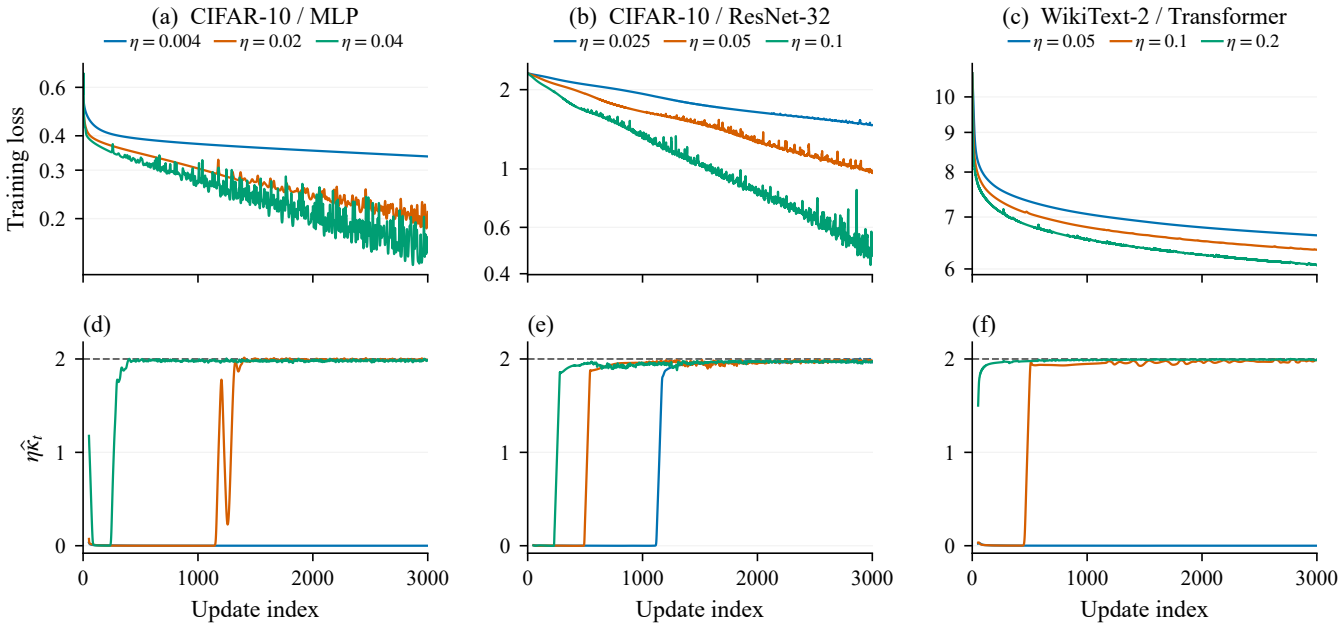



Fig. 1: Training loss and projected gradient change under full-batch GD. Columns identify the dataset/model; colors identify learning rates within each column. Top: unsmoothed full-objective loss on logarithmic axes. Bottom: complete 50-pair trailing means of the pointwise $\eta\widehat{\kappa}_t$, with the local quadratic loss-change boundary at two. Loss at index $t$ is evaluated at $\theta_t$; gradient diagnostics use $g_t$ at $\theta_{t-1}$. Each model uses one shared initialization. Trailing means summarize sustained regimes.

Together, $\rho_t$, gradient norms and $R_t$ distinguish local overshoot, step size and retained motion. Their joint interpretation explains how oscillatory training continues to make objective progress.

### C. Oscillation Beyond Adjacent Updates

To examine the temporal extent of reversal, we fix one endpoint gradient and compare it with preceding gradients at every lag $k = 1, \ldots, 256$. Figure 3 uses the largest tested rate for each model and the common endpoint after 2,256 updates. Strongly opposing and reinforcing pairs recur at longer separations, including after intermediate lags with near-zero similarity. The varying amplitude and sign pattern show that oscillation extends beyond a single adjacent pair, while its persistence depends on the model and part of the trajectory. Adjacent cosine identifies the local regime within this longer temporal structure; cumulative retention summarizes its net geometric effect.

### D. How Sampling Changes the Regime

We vary batch size at initialization and after 2,000 GD updates in the same three model–dataset cases, holding $\eta = 0.02$, $0.1$ and $0.2$ fixed for the MLP, ResNet-32 and Transformer, respectively. Each starting state receives 200-update continuations, with three initializations, eight sampling repetitions per stochastic condition and one full-GD control per initialization. Figure 4 compares mean sampled consecutive-gradient cosine over each continuation. Equal update budgets imply unequal data exposure; sampling units, objective weighting and numerical scope are in Appendix B.

The smallest batches weaken the trained-state destructive regime in all three cases. At initialization, full GD gives a constructive mean for MLP and ResNet, while the Transformer develops a strongly destructive regime within the first 200 updates. The MLP's sampled paths can instead develop negative mean alignment. Sampling therefore influences the regime by changing both the observed gradients and the states that subsequent updates visit.

For the Transformer at $b = 8$, both starting states have mean cosine near $-0.26$, yet mean full-data loss falls from initialization and rises after GD training. The MLP's smallest trained-state batch likewise raises mean full-data loss, whereas full GD lowers it in both models. These outcomes reinforce the role of the joint diagnostics: cosine identifies the directional regime, and gradient magnitudes and loss reveal its consequences. The next section examines this distinction under Adam, where temporal averaging and adaptive scaling further transform the gradient before each step.

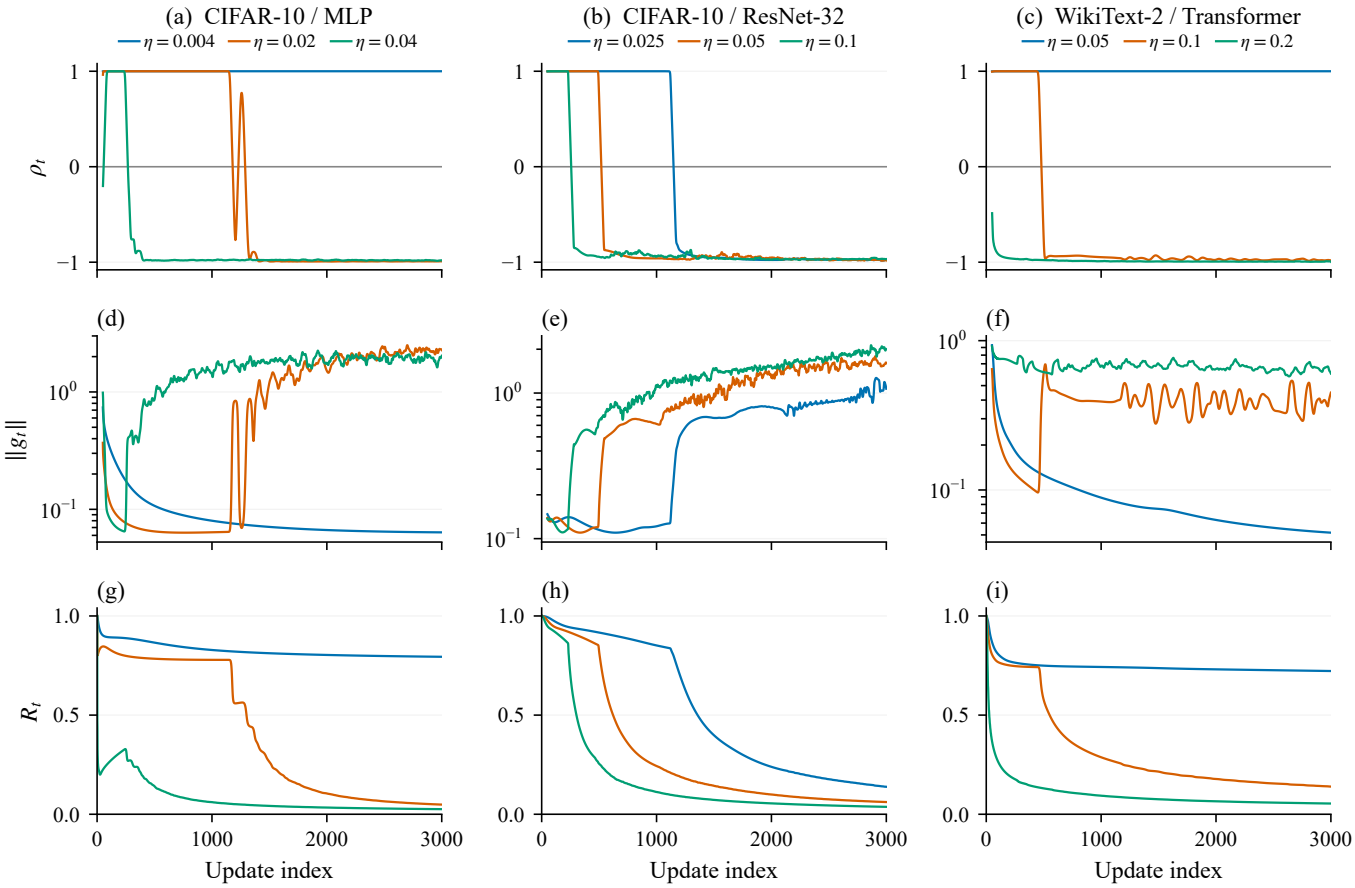



Fig. 2: Directional coherence and cumulative motion for the conditions in Fig. 1. Rows show consecutive cosine, gradient norm (logarithmic scale), and normalized displacement. Cosines and norms use complete 50-observation trailing means; $R_t$ is the unsmoothed ratio over the complete prefix through update $t$. Colors and column ordering match Fig. 1.

TABLE I: Endpoints at $T = 3000$ updates. The late means $\overline{\rho}_{\text{late}}$ and $\overline{\eta\widehat{\kappa}}_{\text{late}}$ average gradient indices $t = 2001, \ldots, 3000$. Displacement is $\|\Delta_T\| = \eta \|\sum_{t=1}^{T} g_t\|$, subject to parameter-update roundoff. Each model shares one initialization across rates; comparisons are within models. *Positive value below 0.00005, rounded to zero.

| Dataset | Model | $\eta$ | $\mathcal{L}(\theta_T)$ | $\overline{\rho}_{\text{late}}$ | $R_T$ | $\overline{\eta\widehat{\kappa}}_{\text{late}}$ | $\|\Delta_T\|$ |
|---|---|---|---|---|---|---|---|
| CIFAR-10 | MLP | 0.004 | 0.3365 | 1.0000 | 0.7941 | 0.0000* | 0.9511 |
| | | 0.02 | 0.2114 | -0.9913 | 0.0488 | 1.9924 | 3.1391 |
| | | 0.04 | 0.1539 | -0.9799 | 0.0257 | 1.9820 | 4.7628 |
| | ResNet-32 | 0.025 | 1.4665 | -0.9723 | 0.1385 | 1.9740 | 5.7045 |
| | | 0.05 | 0.9672 | -0.9708 | 0.0616 | 1.9746 | 9.0183 |
| | | 0.1 | 0.4898 | -0.9684 | 0.0374 | 1.9707 | 13.9329 |
| WikiText-2 | Transformer | 0.05 | 6.6301 | 1.0000 | 0.7223 | 0.0002 | 11.3183 |
| | | 0.1 | 6.3519 | -0.9749 | 0.1395 | 1.9754 | 15.6026 |
| | | 0.2 | 6.0697 | -0.9937 | 0.0541 | 1.9937 | 21.9693 |

## V. Diagnosing Loss Spikes under Adam

Adam tests whether curvature information recovered from training gradients remains useful along adaptive steps. The diagnostics in Eqs. (8) and (15) monitor the regime and projected gradient change using the same one-gradient buffer. The raw gradients reveal the encountered loss geometry while also supplying the input to Adam's adaptive update.

Adam [11] averages this input through first and second moments,

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^{\odot 2}, \end{aligned} \tag{24}$$

starting from zero. Bias correction gives $\widehat{m}_t = m_t/(1-\beta_1^t)$

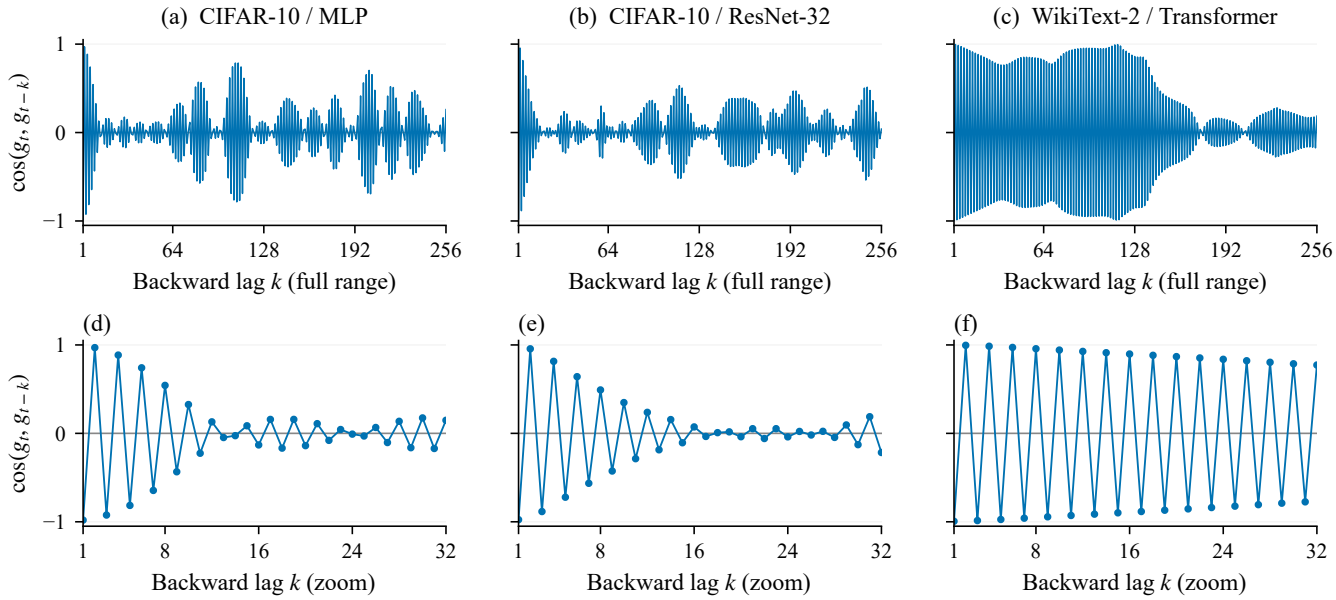



Fig. 3: Backward gradient alignment at the common endpoint state $\theta_{2256}$, corresponding to $t = 2257$ under the manuscript's preupdate gradient indexing. Each point is one exact cosine $\cos(g_t, g_{t-k})$; the endpoint is fixed across all 256 lags. The top row shows lags 1–256; the bottom row magnifies the same values at lags 1–32 to resolve the early sign alternation. Rates are 0.04 (MLP), 0.1 (ResNet-32) and 0.2 (Transformer). These are separately recomputed 256-update GD continuations from retained state 2000, with one initialization and no smoothing or endpoint averaging.

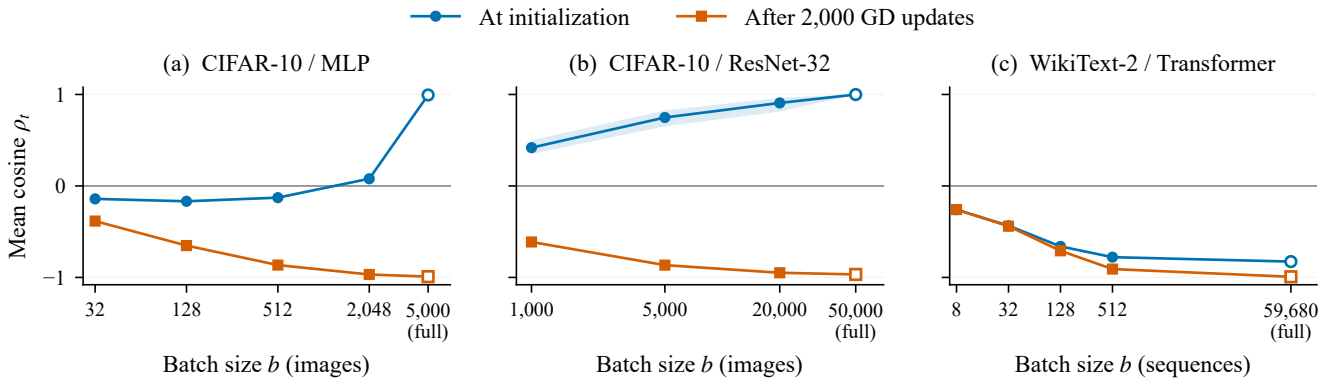



Fig. 4: Mean sampled-gradient cosine over 199 adjacent pairs during 200-update continuations in three model–dataset cases. Blue circles start at initialization; orange squares start after 2,000 GD updates. Each line averages repetitions within initialization, then weights the three initialization blocks equally; shading is their range, not a confidence interval. Hollow rightmost markers denote full GD. ResNet batches contain whole fixed 1,000-image normalization groups; Transformer batches count prepared sequences. Batch axes are logarithmic; lines guide the eye between measured sizes.

and $\widehat{v}_t = v_t/(1-\beta_2^t)$, followed by

$$d_t = \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon}, \qquad \theta_t = \theta_{t-1} - \eta d_t, \tag{25}$$

with coordinatewise operations. The first moment carries information from preceding gradients, while the second moment scales coordinates. A sharp change in the raw gradient therefore enters an optimizer whose response also depends on its accumulated history.

The connection to the loss landscape follows from the same local expansion used for GD. For full-batch gradients along Adam's step,

$$\begin{aligned} \eta\widehat{\kappa}_t &= -\frac{g_{t-1}^\top (g_t - g_{t-1})}{\|g_{t-1}\|^2} \\ &\approx \eta\, \frac{g_{t-1}^\top H_{t-1} d_{t-1}}{\|g_{t-1}\|^2}. \end{aligned} \tag{26}$$

Here the Hessian acts on the adaptive direction, so the diagnostic measures a projected curvature response along that displacement. The path-average version is exact on a smooth step (Appendix A); the Rayleigh quotient in Eq. (12) applies when the step follows the raw gradient. Both responses are

observed without evaluating the Hessian. The exact expression $1 - \rho_t \|g_t\|/\|g_{t-1}\|$ separates that response into the current regime and relative gradient magnitude. In a destructive regime, an increase in the opposing gradient component raises $\eta\widehat{\kappa}_t$; exceeding two means that component is larger than the preceding gradient. The formal GD loss boundary motivates this reference, while the Adam experiments below establish its empirical relation to developing spikes.

### A. Controlled Training and Loss Overview

We use the CIFAR-10/MLP, CIFAR-10/ResNet-32 and WikiText-2/Transformer objectives from Section IV. Each trajectory starts from the same frozen, untrained seed-0 weights as its GD counterpart, with fresh Adam moments. Each model receives 1,024 full-objective updates at three constant rates: $\{10^{-4}, 3\times10^{-4}, 10^{-3}\}$ for the MLP and $\{3\times10^{-4}, 10^{-3}, 3\times10^{-3}\}$ for ResNet and Transformer. We use $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. Full-batch accumulation, fixed rates and the absence of clipping or weight decay isolate the diagnostic patterns within the adaptive training dynamics. Appendix B-C gives the numerical and measurement details.

Figure 5 shows sustained loss reduction interrupted by transient spikes, especially at larger rates. The highest tested rate attains the lowest final training loss in every model. The practical problem is therefore to understand and potentially attenuate the excursions within an otherwise productive trajectory. We begin with individual spikes to resolve their temporal structure, then examine their recurrence across all nine runs.

### B. Early Signs and Anatomy of Individual Spikes

For each model, we select the largest recovered relative loss spike at its highest tested learning rate. This loss-based rule selects peaks at states 703, 422 and 794 for MLP, ResNet and Transformer, respectively. Figure 6 shows their raw-gradient and momentum dynamics at individual-update resolution.

All three spikes develop from a destructive raw-gradient regime. As the loss rises, gradient norms grow sharply and $\eta\widehat{\kappa}$ exceeds two before the peak. The momentum norm also grows, showing that the amplified gradients enter Adam's temporal average. The joint view explains the diagnostic: negative $\rho$ identifies the reversal, and the growing norm makes that reversal increasingly strong.

The Transformer gives a particularly clear early warning. At state 788, $\eta\widehat{\kappa}$ first exceeds two while the loss is only 0.40% above its pre-rise level. Two updates later the increase exceeds 10%; the loss reaches its maximum six updates after the diagnostic crossing, at state 794. Gradient and momentum norms peak at states 792 and 793. The diagnostic exposes the developing excursion before the pronounced rise, providing a concrete interval in which an intervention could act.

Recovery reveals how the regime changes after amplification. In the displayed ResNet and Transformer events, the gradients enter constructive segments as projected change falls and loss declines smoothly. The MLP retains a predominantly destructive raw-gradient regime and alternating momentum alignment, with repeated rebounds during recovery. Both patterns reduce the spike, but through different directional dynamics: constructive descent in the former examples and shrinking oscillation in the MLP.

These cases connect a visible loss spike to an earlier change in the gradient field traversed by Adam. Their shared feature is amplification within a destructive regime; their different recovery paths show why monitoring the regime and magnitude together is informative throughout the event. We next test how broadly this anatomy recurs.

### C. Defining and Measuring Recurrent Spikes

A spike is an excursion above a fixed pre-rise loss: the loss leaves that level, reaches a maximum, and eventually returns. Keeping the reference fixed treats a rise with several local maxima as one event until recovery. This gives each event a clear beginning, peak and duration using the loss alone.

Write $\mathcal{L}_s = \mathcal{L}(\theta_s)$ for the loss after $s$ updates. For event $i$, let $a_i$ denote the state immediately before its rise, $p_i$ its highest-loss state, and $c_i$ its first return to the pre-rise level. Starting from $c_0 = 0$, define

$$\begin{aligned} a_i &:= \min\{s : c_{i-1} \le s < T,\ \mathcal{L}_{s+1} > \mathcal{L}_s\}, \\ c_i &:= \min\{s : a_i < s \le T,\ \mathcal{L}_s \le \mathcal{L}_{a_i}\}, \\ p_i &:= \min \arg\max_{a_i < s < c_i} \mathcal{L}_s. \end{aligned} \tag{27}$$

The first elevated loss occurs at $a_i + 1$, and the earliest maximum defines $p_i$ if there is a tie. We retain events with observed recovery by $T = 1024$ and resume the search at $c_i$; the search ends when a rise or a subsequent return is absent. All measured pre-rise losses are positive. The peak increase is $100(\mathcal{L}_{p_i}/\mathcal{L}_{a_i} - 1)$ percent; rise and recovery times are $p_i - a_i$ and $c_i - p_i$ updates. Relative to the same baseline, the loss ratios before, at the peak and at recovery are 1, $\mathcal{L}_{p_i}/\mathcal{L}_{a_i}$ and $\mathcal{L}_{c_i}/\mathcal{L}_{a_i} \le 1$.

The distributions in Fig. 7 show that the selected cases belong to a recurring pattern. Larger learning rates extend the distribution toward more severe excursions, while small fluctuations remain common. We focus the anatomy analysis on increases strictly greater than 10%, an amplitude criterion that selects pronounced departures from the local training loss. This yields 129 recovered spikes; the full density places their amplitudes in context.

Table II exposes the cost of these excursions. Both mean and maximum spike increase grow with learning rate among the eligible rates, reaching a maximum of 859% in ResNet. Recovery is substantially slower than the rise in ResNet (17–19 versus 6–8 updates) and Transformer (24–31 versus about 6). A short amplification therefore interrupts progress for a much longer interval. MLP rise and recovery times are more comparable, consistent with its repeated oscillatory pattern.

The regime and norm statistics identify the recurring anatomy. Pre-rise $\rho$ is negative in 117 of 127 defined events, and gradient norms increase from pre-rise to the loss peak in all 129. The diagnostic $\eta\widehat{\kappa}$ is already above two before the rise in 21 of 127 defined cases; its event maximum precedes the loss peak in 82 of 129 events. Thus the most consistent population signature is a destructive initial regime followed by

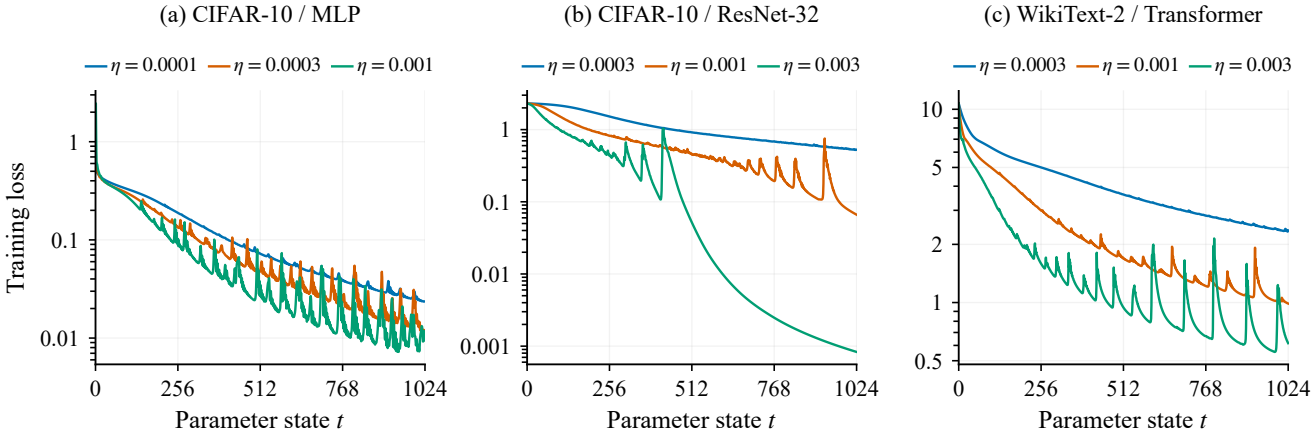



Fig. 5: Training loss under full-batch Adam at three constant learning rates per model. Rates share one frozen initialization within each model. Every loss observation is shown without smoothing on logarithmic y-axes. Parameter state $s$ denotes $s$ completed updates, from initialization through update 1,024. Colors identify the rates in each panel.

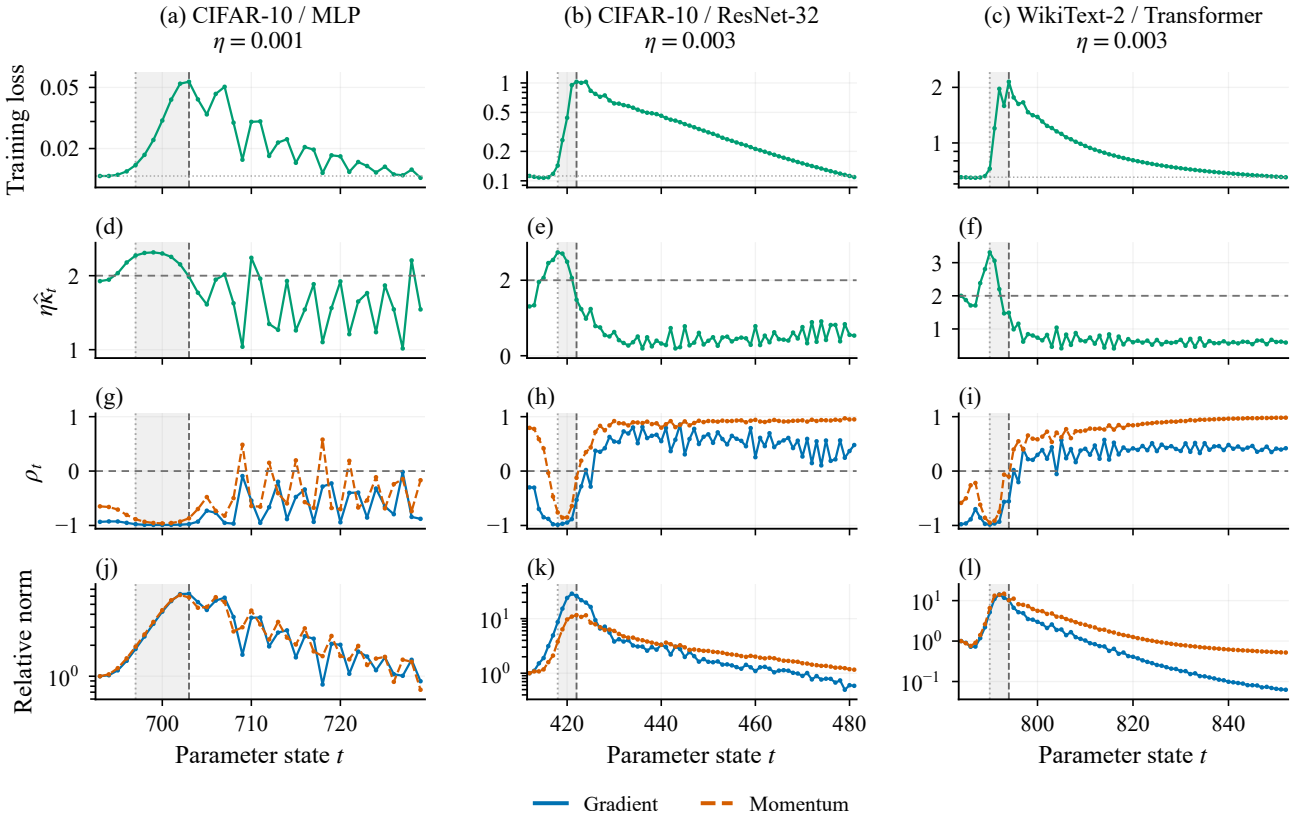



Fig. 6: Largest recovered relative loss spike at each model's highest Adam rate. Rows show training loss, raw-gradient projected change, adjacent cosines of raw gradients (blue) and first moments (orange), and their relative norms. All values are unsmoothed. At parameter state $s$, these diagnostics use $g_{s+1}$ and $m_{s+1}$. Gray bands mark the pronounced rise from the first +10% crossing to the loss peak: 697–703, 418–422 and 790–794. Norms are normalized by their first displayed value. The dotted loss reference is the first displayed loss; windows begin ten states before the peak and end at the first later loss below that reference.

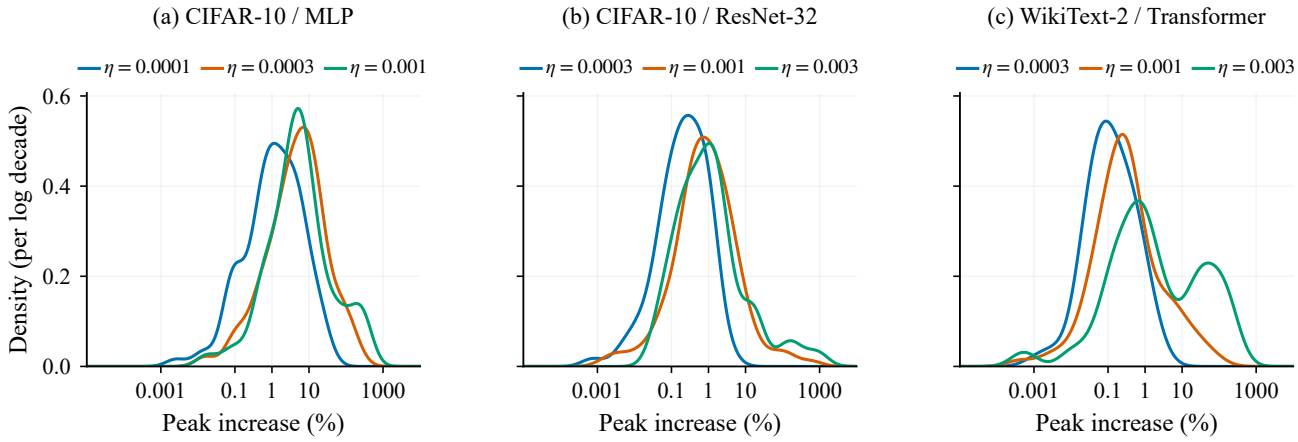



Fig. 7: Peak-increase distributions for all 806 recovered loss excursions. Colors identify learning rates within each model. Gaussian kernel densities use $\log_{10}$ percentage-increase coordinates and a common pooled bandwidth within each model. Each rate's density integrates to one in that coordinate. The logarithmic x-axis shows percentage increase: 1 denotes +1% and 100 denotes +100%.

TABLE II: Anatomy of recovered loss spikes with peak increase $> 10\%$. $\eta$ is the constant Adam learning rate and $N$ the number of events. States $a_i$, $p_i$ and $c_i$ mark pre-rise, peak and first recovery to the pre-rise loss. Entries are event means $\pm$ sample SD, except $N$ and the maximum. Peak increase is $100(\mathcal{L}_{p_i}/\mathcal{L}_{a_i} - 1)$; rise and recovery times are $p_i - a_i$ and $c_i - p_i$ updates. Raw denotes gradients of the training objective before Adam's transformations: $\rho_t = \cos(g_{t-1}, g_t)$ and $\eta\widehat{\kappa}_t = 1 - \rho_t\|g_t\|/\|g_{t-1}\|$. Before uses the single observation at $\theta_{a_i}$ (gradient index $a_i + 1$); recovery summarizes each event's mean over states $p_i+1, \ldots, c_i$. Gradient norm ratios are $\|g_{p_i+1}\|/\|g_{a_i+1}\|$ during the rise and $\|g_{c_i+1}\|/\|g_{p_i+1}\|$ during recovery. Dashes indicate no qualifying event. Two initialization events have undefined before diagnostics, giving 46 and 38 observations for those entries at the MLP's middle and highest rates.

| Dataset Model | $\eta$ | $N$ | Peak increase (%) | | Time (updates) | | $\eta\widehat{\kappa}$ (raw) | | $\rho$ (raw) | | Gradient norm ratio | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Mean $\pm$ SD | Max | Rise | Recovery | Before | Recovery | Before | Recovery | Rise | Recovery |
| CIFAR-10 MLP | $10^{-4}$ | 8 | $19 \pm 6.1$ | 31 | $8.4 \pm 2.9$ | $4.4 \pm 2.4$ | $1.8 \pm 0.38$ | $1.7 \pm 0.08$ | $-0.77 \pm 0.31$ | $-0.89 \pm 0.08$ | $5.3 \pm 1.7$ | $0.30 \pm 0.07$ |
| | $3 \times 10^{-4}$ | 47 | $43 \pm 46$ | 203 | $5.5 \pm 4.4$ | $4.9 \pm 4.0$ | $1.3 \pm 0.25$ | $1.6 \pm 0.18$ | $-0.53 \pm 0.29$ | $-0.71 \pm 0.24$ | $4.5 \pm 3.0$ | $0.33 \pm 0.13$ |
| | $10^{-3}$ | 39 | $95 \pm 107$ | 318 | $6.1 \pm 5.9$ | $7.4 \pm 7.5$ | $1.4 \pm 0.40$ | $1.4 \pm 0.26$ | $-0.46 \pm 0.38$ | $-0.55 \pm 0.30$ | $5.2 \pm 3.8$ | $0.31 \pm 0.14$ |
| CIFAR-10 ResNet-32 | $3 \times 10^{-4}$ | 0 | – | – | – | – | – | – | – | – | – | – |
| | $10^{-3}$ | 10 | $105 \pm 182$ | 602 | $7.5 \pm 2.2$ | $17 \pm 11$ | $1.3 \pm 0.39$ | $1.3 \pm 0.25$ | $-0.36 \pm 0.42$ | $-0.36 \pm 0.26$ | $4.3 \pm 2.9$ | $0.43 \pm 0.24$ |
| | $3 \times 10^{-3}$ | 7 | $179 \pm 309$ | 859 | $6.3 \pm 1.3$ | $19 \pm 20$ | $1.7 \pm 0.38$ | $1.1 \pm 0.37$ | $-0.65 \pm 0.28$ | $-0.13 \pm 0.42$ | $5.0 \pm 4.2$ | $0.34 \pm 0.24$ |
| WikiText-2 Transformer | $3 \times 10^{-4}$ | 0 | – | – | – | – | – | – | – | – | – | – |
| | $10^{-3}$ | 6 | $33 \pm 25$ | 81 | $6.3 \pm 1.9$ | $24 \pm 9.8$ | $2.1 \pm 0.27$ | $0.96 \pm 0.16$ | $-0.84 \pm 0.10$ | $-0.02 \pm 0.12$ | $9.0 \pm 3.8$ | $0.09 \pm 0.05$ |
| | $3 \times 10^{-3}$ | 12 | $82 \pm 70$ | 231 | $6.2 \pm 2.7$ | $31 \pm 17$ | $2.2 \pm 0.43$ | $0.91 \pm 0.19$ | $-0.79 \pm 0.22$ | $0.09 \pm 0.21$ | $11 \pm 6.9$ | $0.08 \pm 0.08$ |

gradient amplification, with projected change often responding ahead of peak loss. During recovery, mean gradient norm ratios fall below one in every group. The MLP and ResNet retain negative group-mean recovery cosines, while Transformer means approach zero: the population also exhibits recovery through shrinking destructive oscillations.

### *D. Regime, Amplification and Spike Magnitude*

The preceding results suggest that spike severity depends jointly on the regime before the rise and the gradient amplification during it. We summarize this combination by plotting

$$\rho_{a_i+1}\frac{\|g_{p_i+1}\|}{\|g_{a_i+1}\|} \quad \text{against} \quad \frac{\mathcal{L}_{p_i}}{\mathcal{L}_{a_i}}. \tag{28}$$

The horizontal coordinate preserves the sign of the pre-rise regime and weights it by the rise norm ratio. More negative values combine stronger initial opposition with larger subsequent amplification. It is a retrospective event descriptor: Eq. (15) instead uses an adjacent norm ratio to describe the response at each update.

Figure 8 shows a clear association between this joint descriptor and spike magnitude. Separate fits at each learning rate explain 67–69% of variation in log relative peak loss for the MLP, 90–94% for ResNet, and 38–54% for the Transformer. The strongest relation occurs in ResNet, while the Transformer retains more variation beyond the two-factor description. These results extend the micro-plots: the destructive regime and gradient amplification jointly organize the severity of

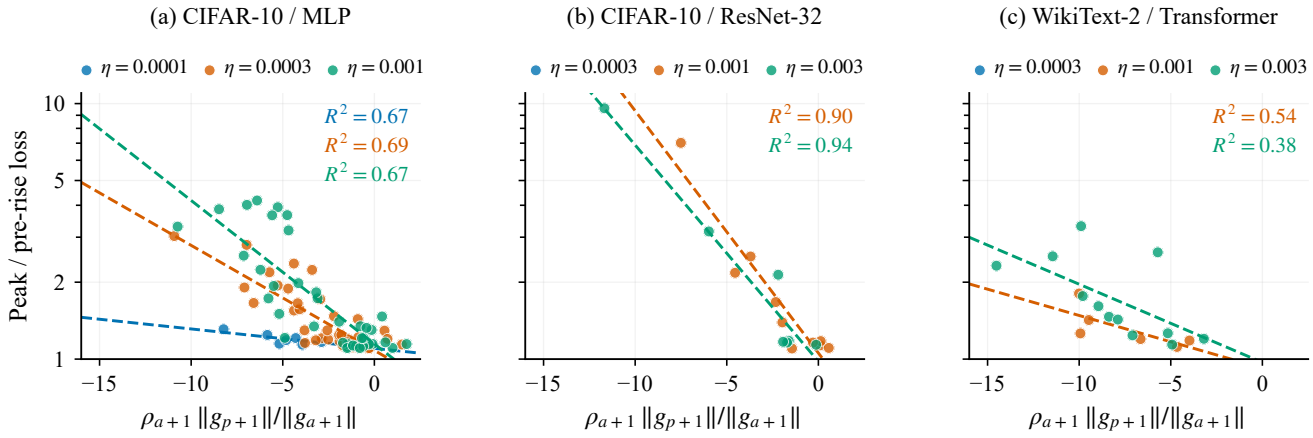



Fig. 8: Pre-rise raw-gradient cosine multiplied by the rise norm ratio versus relative peak loss, as defined in Eq. (28). Each point is one of 127 recovered spikes above 10% with defined pre-rise cosine. Colors identify learning rates. Dashed per-rate least-squares lines fit $\log_{10}$(peak/pre-rise loss) with an intercept and extend across the panels. Color-matched $R^2$ values measure explained variation on that log response in the fitted sample. The logarithmic y-axis shows a loss ratio: 1 denotes no increase and 2 denotes +100%.

repeated spikes across the three cases.

This anatomy connects curvature-related instability to familiar stabilization mechanisms. Learning-rate reduction shortens the step relative to the traversed curvature; clipping caps the optimizer's input norm; momentum averages changing directions; data mixing changes both their alignment and sampling variability. Gradient clipping directly controls the large input norms implicated here, consistent with its use against exploding gradients [19] and theoretical accounts of adaptivity under gradient-dependent smoothness [20]. In Adam it also changes both moment streams. The diagnostics expose which component changes during a spike, providing a mechanistic hypothesis for why a stabilization choice helps and when it may be useful. Their early response motivates testing interventions before the pronounced rise against unchanged continuations from the same saved state.

The present evidence establishes temporal warning signs and within-run associations in three controlled model–dataset cases. Events share trajectories, and the scatter includes gradient norms at the eventual loss peak. Prospective detection and causal tests are therefore the next stage: use only information available at the decision state, include non-spike periods, and measure both spike attenuation and subsequent loss at matched training budgets.

## VI. Conclusion

Consecutive training gradients make directional curvature observable without Hessian computation. For smooth full-batch GD, the estimate equals curvature averaged along the step and approximates its starting-point value; one retained gradient and linear-time reductions support online monitoring. Adjacent cosine identifies constructive and destructive regimes, and the norm ratio expresses the amplification entering the estimate. The local loss relation explains the distinction: destructive training can sustain descent, while sufficiently amplified reversal produces loss-increasing steps. Controlled GD and SGD experiments characterize these regimes and their cumulative displacement.

Under Adam, the same measurements reveal projected curvature response along adaptive trajectories. Diagnostic changes precede the largest loss peaks, and the event analysis connects spike magnitude to a predominantly destructive initial regime followed by gradient amplification. The micro-plots supply temporal evidence, while the distributions and regressions show how the pattern repeats across models and rates. Together, they identify observable early signs and a consistent anatomy of local loss instability.

This connection between curvature, regime and amplification provides an interpretable basis for stabilization heuristics. Learning-rate adjustment, clipping and momentum act on the step or gradient dynamics exposed by the diagnostics. Prospective warning rules and matched intervention tests can turn this online information into decisions that attenuate spikes while preserving training progress.

## Appendix A
## Curvature Estimation and Local Loss Change

Write $H(\theta) = \nabla^2 \mathcal{L}(\theta)$ and assume the objective is twice continuously differentiable along the plain-SGD step. Integrate its gradient derivative on $\theta_{t-2} - s\eta g_{t-1}$, add the endpoint-noise difference, and project onto $-g_{t-1}/(\eta\|g_{t-1}\|^2)$. The result is

$$\widehat{\kappa}_t = \int_0^1 \frac{g_{t-1}^\top H(\theta_{t-2} - s\eta g_{t-1}) g_{t-1}}{\|g_{t-1}\|^2}\, ds - \frac{\varepsilon_t}{\eta}. \tag{29}$$

If $H$ is $L_H$-Lipschitz in operator norm on the step, its full-batch specialization satisfies

$$|\widehat{\kappa}_t - \kappa_t| \le \int_0^1 L_H s\eta \|g_{t-1}\|\, ds = \frac{L_H \eta}{2}\|g_{t-1}\|. \tag{30}$$

This bounds the difference between the measured path average and the starting-point directional curvature. For Adam, integrate along its adaptive step and project onto the preceding raw gradient. This gives the exact path-average form of Eq. (26).

Consider $\theta_{t-1} = \theta_{t-2} - \eta g_{t-1}$ and write $\delta\mathcal{L}_{t-1} = \mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$. Integrating the full gradient along the step gives

$$\delta\mathcal{L}_{t-1} = -\eta \int_0^1 g_{t-1}^\top \nabla\mathcal{L}(\theta_{t-2} - s\eta g_{t-1})\, ds. \tag{31}$$

The trapezoidal approximation replaces the integrand by its endpoint average. Substituting $\nabla\mathcal{L}(\theta_{t-2}) = g_{t-1} - \xi_{t-1}$ and $\nabla\mathcal{L}(\theta_{t-1}) = g_t - \xi_t$ yields

$$\begin{aligned}\delta\mathcal{L}_{t-1} \approx -\frac{\eta}{2}\left(\|g_{t-1}\|^2 + g_{t-1}^\top g_t\right) \\ + \frac{\eta}{2} g_{t-1}^\top(\xi_{t-1} + \xi_t).\end{aligned} \tag{32}$$

The substitution $g_{t-1}^\top g_t = \|g_{t-1}\|\|g_t\|\rho_t$ recovers Eq. (21). The expression is exact on quadratic objectives. A locally Lipschitz Hessian bounds its error by a constant times $\|\eta g_{t-1}\|^3$.

Using $\rho_t\|g_t\|/\|g_{t-1}\| \approx 1 - \eta\kappa_t + \varepsilon_t$ and Eq. (13) gives the equivalent curvature form,

$$\begin{aligned}\delta\mathcal{L}_{t-1} \approx -\eta\|g_{t-1}\|^2\left(1 - \frac{\eta\kappa_t}{2}\right) \\ + \eta\xi_{t-1}^\top g_{t-1}.\end{aligned} \tag{33}$$

It also follows directly from a second-order Taylor expansion at $\theta_{t-2}$. With zero noise, the conditions for a loss-increasing step reduce to Eqs. (22) and (23).

The noise term has a simple conditional interpretation. Given $\theta_{t-2}$, unbiased sampling gives $\mathbb{E}[\|g_{t-1}\|^2] = \|\nabla\mathcal{L}(\theta_{t-2})\|^2 + \mathbb{E}[\|\xi_{t-1}\|^2]$ and $\mathbb{E}[\xi_{t-1}^\top g_{t-1}] = \mathbb{E}[\|\xi_{t-1}\|^2]$. Their first-order variance terms cancel, leaving

$$\begin{aligned}\mathbb{E}[\delta\mathcal{L}_{t-1} \mid \theta_{t-2}] \approx -\eta\|\nabla\mathcal{L}(\theta_{t-2})\|^2 \\ + \frac{\eta^2}{2}\mathbb{E}[g_{t-1}^\top H_{t-1} g_{t-1} \mid \theta_{t-2}].\end{aligned} \tag{34}$$

Thus sampling contributes through curvature at second order. A nonlinear objective also contributes the finite-step remainder.

## Appendix B
## Essential Measurement Conventions

### A. Fixed Objectives and Numerical Scope

The tanh MLP has two 200-unit hidden layers and minimizes half-squared error summed over ten outputs and averaged over the first 5,000 CIFAR-10 images. ResNet-32 minimizes mean cross-entropy over all 50,000 training images, using 50 fixed groups of 1,000 for batch normalization. Standardization uses the complete CIFAR-10 training split: coordinate-wise for the MLP and channel-wise for ResNet. Full-batch accumulation evaluates the same fixed objective at every update; augmentation and dropout are disabled.

The word-level Transformer has two layers, two heads and embedding/feed-forward dimensions of 200. Twenty corpus streams form causal chunks of at most 35 tokens, yielding 59,680 sequences and 2,088,600 supervised targets. The objective averages negative log-likelihood over valid targets, excluding padding. Architecture and the 33,278-word first-occurrence dictionary over training, validation and test text follow PyTorch example revision `13acec6d7c78` [21].

Diagnostics cover all trainable parameters. Parameters, gradients and Adam moments use FP32; reductions use FP64, with TF32 disabled. Initial unpaired cosines and zero-denominator ratios are undefined. The GD plots' trailing means summarize sustained regimes and can delay a displayed transition; Adam losses and micro-plots are unsmoothed. All claims concern training objectives. Matched tensors give one initialization per full-batch model; independent hardware executions can develop finite-precision drift.

### B. Sampling and Trajectory Comparisons

Stochastic batches are independent uniform subsets without replacement; within a repetition, batch sizes share prefixes of a common permutation. ResNet samples complete normalization groups. Transformer sampling uses $N = 59{,}680$ sequences and $M = 2{,}088{,}600$ total targets: for sequence summed losses $\ell_i$, the estimator is $N(bM)^{-1}\sum_{i\in B}\ell_i$. Its fixed denominator preserves the full token-weighted objective. The seed-2 smallest two ResNet batches use a different runtime, with 200-update full-GD replay parameter relative error $2.36 \times 10^{-7}$; batch and runtime remain confounded in this block. The backward-lag curves also use separate continuations, whose later floating-point paths can differ from the original GD trajectories.

### C. Adam Event Summaries

At state $\theta_s$, the raw gradient and updated first moment are $g_{s+1}$ and $m_{s+1}$, before parameter update $s+1$. The terminal raw gradient is evaluated without another update. Hardware varies across ResNet and Transformer learning-rate arms; initial-gradient agreement is checked, while subsequent finite-precision drift remains possible. Rate selection uses training calibration.

Two excursions above 10% without observed recovery are excluded. Two retained initialization events have undefined pre-rise diagnostics. Each event has equal weight; recovery diagnostics first average within the event, and sample SD uses denominator $n - 1$. The scatter uses per-rate ordinary least squares with an intercept on log relative peak loss. Its $R^2$ describes the fitting sample; dashed extensions extrapolate. For the density, each model's pooled log-percentage SD times pooled $n^{-1/5}$ sets the common Gaussian bandwidth. Events within a trajectory can be dependent.

## References


[1] J. M. Cohen, S. Kaur, Y. Li, J. Z. Kolter, and A. Talwalkar, "Gradient descent on neural networks typically occurs at the edge of stability," in *International Conference on Learning Representations (ICLR)*, 2021.

[2] A. Chowdhery, S. Narang, J. Devlin, M. Bosma, G. Mishra, A. Roberts *et al.*, "PaLM: Scaling language modeling with pathways," *arXiv preprint arXiv:2204.02311*, 2022. [Online]. Available: https://arxiv.org/abs/2204.02311v5

[3] Team OLMo, P. Walsh, L. Soldaini, D. Groeneveld, K. Lo *et al.*, "2 OLMo 2 furious," *arXiv preprint arXiv:2501.00656*, 2025. [Online]. Available: https://arxiv.org/abs/2501.00656v3

[4] B. T. Polyak, "Some methods of speeding up the convergence of iteration methods," *USSR Computational Mathematics and Mathematical Physics*, vol. 4, pp. 1–17, 1964.

[5] S. Jastrzębski, M. Szymczak, S. Fort, D. Arpit, J. Tabor, K. Cho, and K. Geras, "The break-even point on optimization trajectories of deep neural networks," in *International Conference on Learning Representations (ICLR)*, 2020.

[6] S. Arora, Z. Li, and A. Panigrahi, "Understanding gradient descent on the edge of stability in deep learning," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162. PMLR, 2022, pp. 948–1024.

[7] A. Damian, E. Nichani, and J. D. Lee, "Self-stabilization: The implicit bias of gradient descent at the edge of stability," in *International Conference on Learning Representations (ICLR)*, 2023.

[8] J. Chee and P. Toulis, "Convergence diagnostics for stochastic gradient descent with constant learning rate," in *Proceedings of the 21st International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 84, 2018, pp. 1476–1485. [Online]. Available: https://proceedings.mlr.press/v84/chee18a.html

[9] S. Pesme, A. Dieuleveut, and N. Flammarion, "On convergence-diagnostic based step sizes for stochastic gradient descent," in *Proceedings of the 37th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 119, 2020, pp. 7641–7651. [Online]. Available: https://proceedings.mlr.press/v119/pesme20a.html

[10] A. G. Baydin, R. Cornish, D. Martínez Rubio, M. Schmidt, and F. Wood, "Online learning rate adaptation with hypergradient descent," in *International Conference on Learning Representations*, 2018. [Online]. Available: https://arxiv.org/abs/1703.04782

[11] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.

[12] D. P. Bertsekas, *Nonlinear Programming*, 2nd ed. Belmont, MA: Athena Scientific, 1999.

[13] C. Xing, D. Arpit, C. Tsirigotis, and Y. Bengio, "A walk with SGD," *arXiv preprint arXiv:1802.08770*, 2018.

[14] S. Mandt, M. D. Hoffman, and D. M. Blei, "Stochastic gradient descent as approximate bayesian inference," *Journal of Machine Learning Research*, vol. 18, no. 134, pp. 1–35, 2017. [Online]. Available: https://www.jmlr.org/papers/v18/17-214.html

[15] W. Hu, C. J. Li, L. Li, and J.-G. Liu, "On the diffusion approximation of nonconvex stochastic gradient descent," *Annals of Mathematical Sciences and Applications*, vol. 4, no. 1, pp. 3–32, 2019.

[16] S. Jastrzębski, Z. Kenton, D. Arpit, N. Ballas, A. Fischer, Y. Bengio, and A. Storkey, "Three factors influencing minima in sgd," *arXiv preprint arXiv:1711.04623*, 2017. [Online]. Available: https://arxiv.org/abs/1711.04623

[17] S. Zhang, S. Roller, N. Goyal, M. Artetxe, M. Chen, S. Chen *et al.*, "OPT: Open pre-trained transformer language models," *arXiv preprint arXiv:2205.01068*, 2022. [Online]. Available: https://arxiv.org/abs/2205.01068v4

[18] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle *et al.*, "The Llama 3 herd of models," *arXiv preprint arXiv:2407.21783*, 2024. [Online]. Available: https://arxiv.org/abs/2407.21783v3

[19] R. Pascanu, T. Mikolov, and Y. Bengio, "On the difficulty of training recurrent neural networks," in *International conference on machine learning*. Pmlr, 2013, pp. 1310–1318.

[20] J. Zhang, T. He, S. Sra, and A. Jadbabaie, "Why gradient clipping accelerates training: A theoretical justification for adaptivity," in *International Conference on Learning Representations*, 2020. [Online]. Available: https://openreview.net/forum?id=BJgnXpVYwS

[21] PyTorch contributors, "PyTorch examples: Word language model," GitHub repository, revision `13acec6d7c78`; full commit specified in the URL. [Online]. Available: https://github.com/pytorch/examples/tree/13acec6d7c78dacd5e1fe9b0b4a325e1d39abc15/word_language_model